

Lecture 9: The Quotient Rule; Derivatives of the Other Trigonometric Functions

GOAL: Be able to differentiate functions of the form $\frac{f(x)}{g(x)}$. Use this to derive the derivatives of all 6 trigonometric functions.

Summary:

$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$		
$(\sin x)' = \cos x$	$(\cos x)' = -\sin x$	$(\tan x)' = \sec^2 x$
$(\sec x)' = \sec x \tan x$	$(\csc x)' = -\csc x \cot x$	$(\cot x)' = -\csc^2 x$

